

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
GRADE 7 · GRADE 7 MATHEMATICS

MATHEMATICS 702/2

PAPER 2 November 2021

2 hours

PAPER 2 — STRUCTURED QUESTIONS — CALCULATOR ALLOWED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2021 702/2.

Additional materials: Calculators may be used.

INSTRUCTIONS: Answer all questions. Show working.

1. A school garden is a rectangle 12 m by 8 m. <<
>>(a) Perimeter [2]<<
>>(b) Area [2]<<
>>(c) Fencing costs \$3 per metre. Cost to fence it? [2] **[6]**

2. Anesu has \$35. She buys 7 pens at \$2 each. <<
>>(a) Cost of pens [2]<<
>>(b) Change [2]<<
>>(c) If pens were \$3, could she still buy 7? [2] **[6]**

3. A cube of edge 5 cm. <<
>>(a) Volume [2]<<
>>(b) Surface area [3] **[5]**

4. A pizza is cut into 8 equal slices. Tendai eats 3 slices. <<
>>(a) Fraction eaten [1]<<
>>(b) Fraction left [2]<<
>>(c) If 2 pizzas are shared equally by 5 children, what fraction does each get? [2] **[5]**

5. Marks: 4, 6, 6, 7, 8, 9, 10. <<
>>(a) Mean [3]<<
>>(b) Median [2]<<
>>(c) Range [1] **[6]**

6. An isosceles triangle has base angles 60° each. <<
>>(a) Find the vertex angle. [3]<<
>>(b) What type of triangle is it (acute/obtuse/right)? [2] **[5]**

7. A kombi travels 24 km at 4 km per 10 minutes. How many minutes for the trip if it keeps this speed? **[5]**

8. Name the number of lines of symmetry of: <<
>>(a) a square [2]<<
>>(b) a rectangle that is not a square [2]<<
>>(c) a circle [1] **[5]**

- 9.** On a grid, A is (2, 3) and B is (2, 8).<<
>>(a) Plot-describe AB. [2]<<
>>(b) Length of AB. [2]<<
>>(c) Midpoint of AB. [1] **[5]**
- 10.** A bottle holds 2 litres. A cup holds 250 ml.<<
>>(a) How many cups fill the bottle? [3]<<
>>(b) 3 bottles in ml. [3] **[6]**
- 11.** A plan uses scale 1 cm to 5 m. The classroom is 8 cm long on the plan.<<
>>(a) Real length [3]<<
>>(b) Real width if plan width is 6 cm [3] **[6]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).